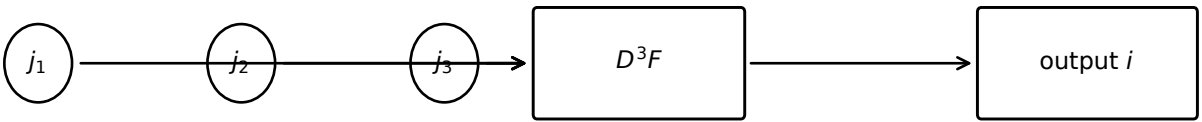
multilinear packet transfer is a typed hyperedge



$$Q_i D^m F(P_{j_1} \cdot, \ldots, P_{j_m} \cdot)$$